

**University Of Michigan – Dearborn**  
**Isolated Bidirectional DC-DC Converter Design**  
**ECE517 - Term Project Report – Fall 2024**

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**Abstract:**

This report presents the design and simulation of a high efficiency isolated bidirectional DC-DC converter used for distributed energy storage systems.

The Dual Active Bridge (DAB) topology has been chosen to achieve bidirectional power transfer between 400 V and 45 V batteries. The design includes the traditional PID control system, to regulate the phase shift angle and improve performance.

MATLAB Simulink simulations confirmed the converter's ability to meet the project requirements, achieving a power efficiency exceeding 95%, voltage ripple below 3%, and current ripple under 4%. These results highlight the potential of the proposed DAB converter for applications in energy storage systems, electric vehicles, and smart grid solutions.

## **Introduction:**

Isolated bidirectional DC-DC converters with high efficiency performance are critical and major components in various electrical system applications, such as energy storage systems, automotive industry, and smart grid applications. These types of converters transfer energy in both directions and provide isolation between the high voltage and low voltage sides typically using transformers, for safety purposes and to improve performance in high voltage applications.

One key application that requires an isolated bidirectional DC-DC converter is distributed energy storage devices, where this type of converter allows batteries on both sides to charge and discharge energy with high efficiency. Isolating the low and high voltage side provides safety for these batteries and the overall system [1].

The project objective is to design a high efficiency isolated bidirectional DC-DC converter for distributed energy storage devices (batteries). The Dual Active Bridge (DAB) converter was used in this project to distribute the energy between 400 V and 45 V batteries, with efficiency higher than 95% at 2kW full load.

The Dual Active Bridge (DAB) converter is one of the common isolated bidirectional DC-DC converters topologies. It is known for its high efficiency and power density [1]. Using high frequency transformers in these converters is required to reduce the size and make it more compact while keeping the efficiency high [5].

The DAB converter consists of two full bridge converters connected through a high-frequency transformer, The power transfers from one side to the other side due to the phase shift between the primary and the secondary side of the transformer. The phase shift can be generated and controlled by regulating the two full bridge converters using pulse width modulation signals (PWM) [10],[11].

In summary, DAB converters with the traditional PID controller are designed in this project to distribute power up to 2kW between 400 V and 45 V batteries, with output ripples less than 3% and efficiency higher than 96%.

### **Design Procedure:**

The project requires designing a high efficiency isolated bidirectional DC-DC converter, to distribute power up to 2kW between 400 V and 45 V batteries, with output current ripple less than 4%, voltage ripple less than 3% at full load and efficiency higher than 95%.

The first step is to design the high and low voltage ports that consist of 400V and 45V batteries with rated power up to 2kW and DC links to filter the noise, store the energy and stabilize the voltage in each side.

The required capacitor dc link can be calculated using *Equation 1*.

$$C \geq \frac{I_{ripple}}{4 f \Delta V} \rightarrow C \geq \frac{1.6}{4 \cdot 50 \cdot 12} = 667\mu F \text{ (Equation 1)}$$

KEMET ALS70A102NF450 1mF capacitor has been chosen as a dc link capacitor with a 1mΩ ESR resistance, it has a maximum voltage higher than 400V [12]. HDL3000PS400 45V battery with 20mΩ internal resistance and RSP-2000-48 45V with 60mΩ internal resistance from Digikey supplier were used in the design [13][14].

Typical type of transistors can be used to design the two full bridges in the DAB converter, the transistor must handle voltage higher than 400V. IPP110N20N3/IPW60R037P7 transistors with 11mΩ transistor on-state resistance, 1MΩ Snubber resistance Rs, 10mΩ and 0.9 diode voltage drop were utilized to build the two full bridges [15][16].

*Table 1* shows the parameters that were chosen to design the 400/45 transformer. Infineon Technologies can design this type of transformer [17].

| Parameter                | Value     | Comments                                                                                                                                                    |
|--------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rated Power              | 2kW       | This value was given by the project's requirements. However, higher power rate is preferred.                                                                |
| Switching Frequency      | 50 k (Hz) | Frequencies around 50–100 kHz are common, balancing efficiency and transformer size.                                                                        |
| Primary side resistance  | 10mΩ      | The system works fine even if the resistance is increased up to 50m Ohm.                                                                                    |
| leakage inductance       | 15μH      | Transformer with leakage inductance between 10-15 μH is common [15]                                                                                         |
| Magnetization resistance | 10mΩ      | Resistance can be increased to reduce the magnetization current. However, after a certain value the loss can be increased because of high resistance value. |
| Magnetization inductance | 16mH      | Common in transformers [15]                                                                                                                                 |

Table 1: transformer parameters

The suitable DAB coupling inductance can be calculated using *Equation 2*, which represents the inductance that is required to generate the maximum power, such an inductance can be found in different supplier websites. 744355110 is an example of 100μH inductance with a 4.5mΩ DCR resistance.

$$L \geq \frac{V_1 V_2}{2 \pi f P_{max}} \rightarrow L \geq \frac{400*45}{2 \pi 50k 2k} = 30\mu H. (Equation 2)$$

*Figure 1* shows the control system that was used to control the DAB converter. The control system consists of PID controllers, and a pulse generator which generates PWM signals that control the full bridge gates. Based on the difference between the actual and the reference output power, a phase shift angle between the two sides of the transformer will be calculated and added to the PWM signals to increase or decrease the power transfer trying to minimize the error signal.

There are various methods for tuning PID controllers, such as Ziegler-Nichols, Cohen-Coon, and trial-and-error methods. For the Ziegler-Nichols method:

- Start with small values (e.g.,  $K_p=0.1$  and  $K_i=0.01$ ).
- Increase  $K_p$  until you observe oscillations in the step response.
- Increase  $K_i$  until you achieve the desired performance.
- Adjust the controller settings based on how well the system performs and what the system needs.

Using this method the proportional gain was selected to be 0.1 and the integral gain is 600.

Higher values will make the steady state error better but the ripple worse.

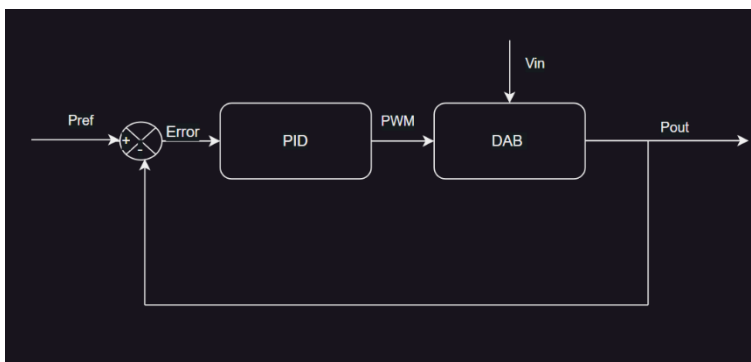


Figure 1: Control system design

## Simulation Model:

MATLAB Simulink was used to simulate the DAB converter, using the parameters and variables that are listed in Table 2. All these values were selected based on the calculations and the justifications mentioned in the previous section.

| Parameter         | Value                        |
|-------------------|------------------------------|
| $V_{s1}$          | 400 V                        |
| $V_{s2}$          | 45 V                         |
| $R_1$             | 20 m $\Omega$                |
| $R_2$             | 60 m $\Omega$                |
| DC Link Capacitor | 100 $\mu$ F + 1 m $\Omega$   |
| $P_n$             | 2 kW                         |
| $f_s$             | 50 kHz                       |
| $T_s$             | 1 $\mu$ s                    |
| $R_{prim}$        | 10 m $\Omega$                |
| $L_{prim}$        | 15 $\mu$ H                   |
| $R_s$             | 1 $\mu\Omega$                |
| $R_m$             | 10 k $\Omega$                |
| $L_m$             | 16 mH                        |
| $R_{on}$          | 11 m $\Omega$                |
| $R_{sn}$          | 1 M $\Omega$                 |
| $C_{sn}$          | $\infty$                     |
| $R_d$             | 10 m $\Omega$                |
| $V_d$             | 0.9 V                        |
| Coupling Branch   | 100 $\mu$ H + 4.5 m $\Omega$ |

Table 2: Simulation Model parameters and values

The Simulink model consists of High voltage port, Low voltage port includes the 45 V battery and its DC links, and three subsystems (DAB converter, Microcontroller and the sensing system).

As shown in *Figure 2* the high voltage and low voltage ports contain the 400V and 45 Batteries, and their DC link branches.

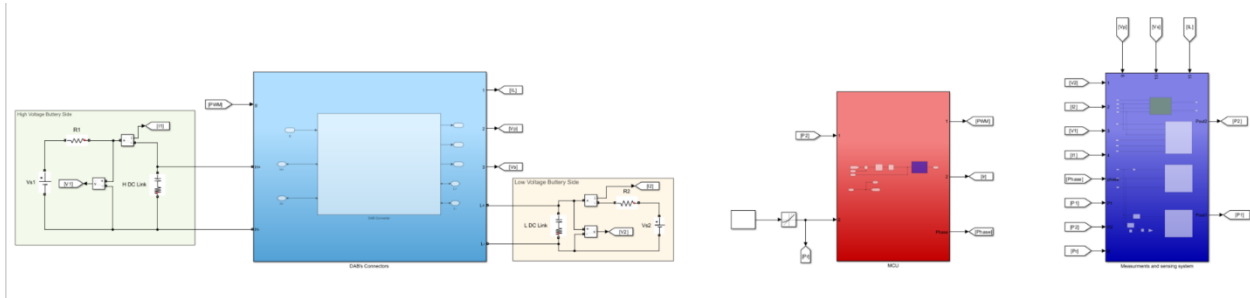


Figure 2: Top Level Schematic

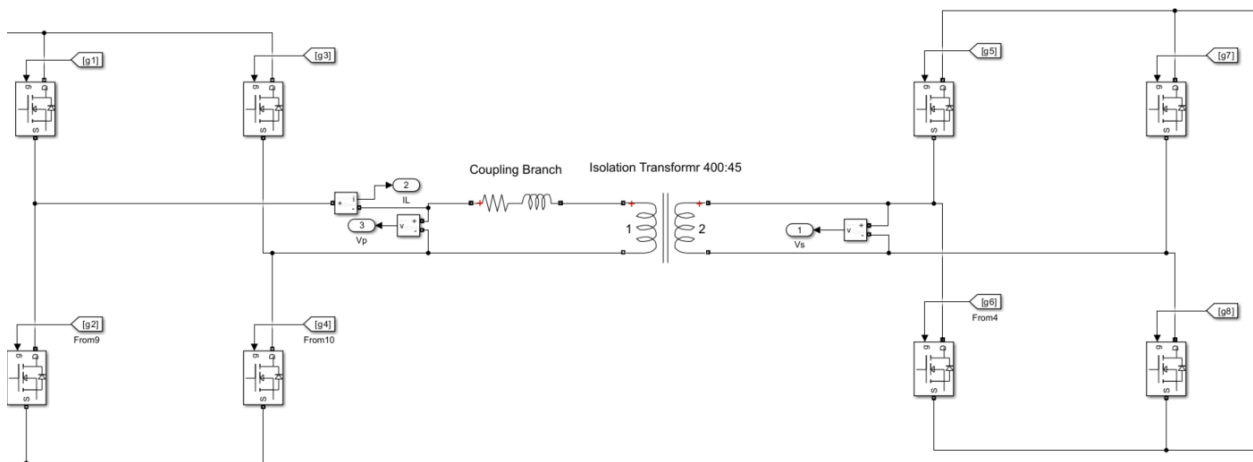


Figure 3: DAB converter



The DAB converter subsystem consists of two full bridges, the coupling inductor and the 400/45 transformer. Gate 1 and Gate 4 must have the same PWM signal to work simultaneously and provide a path to +400 V while Gate 2 and Gate 3 provide a path to -400 V. The same concept applies on the second full bridge. By controlling the full bridge gates, a phase shift between the primary side and the secondary side voltage will be added and the power will be transferred from the leading side to the lagging one according to *Equation 3*.

$$Power = \frac{V_1 V_2}{X_L} f(\text{phase shift}) \quad (\text{Equation 3})$$

When high voltage source is discharging the power transfers from the primary side to secondary side, which means the primary side voltage leads the secondary side voltage by certain phase shift depending on the power value. In contrast when the high voltage side charges the energy transfers from the secondary to the primary and the secondary side voltage leads the primary side voltage by the phase shift angle as shown in *Figure 4* and *Figure 5*.

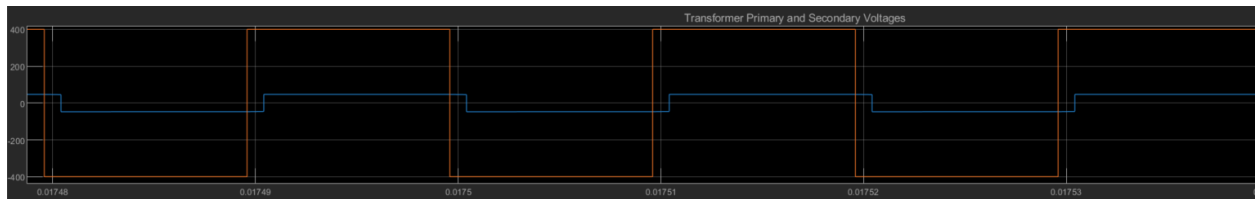


Figure 4: Transformer primary and secondary voltage during discharging energy from the high voltage side to the low voltage side

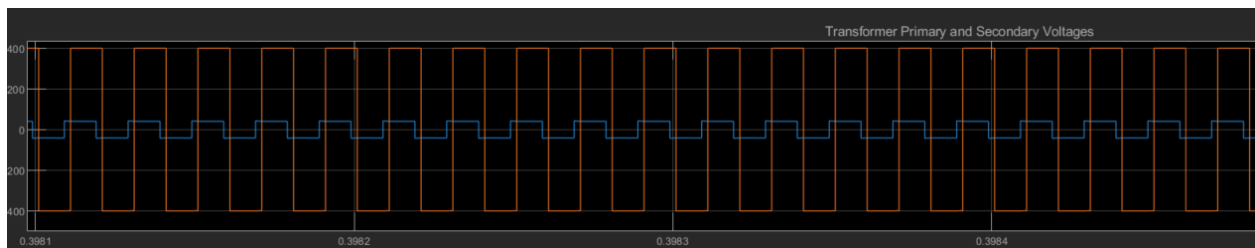


Figure 5: Transformer primary and secondary voltage during discharging energy from the low voltage side to the high voltage side

The controller system was represented in the microcontroller (MCU) subsystem, the MCU takes the output power and compares it with the references power that was chosen to be a signal that goes up from zero to 2kW to represent source 2 charging statuses, then it goes from 2kW to -2kW to represent the discharging statuses. The difference between the actual power and the references power multiplied by PID parameters to improve the steady state error and the performance represents the required phase shift angle, the phase shift goes to saturation block to make sure it is between -35 degrees to prevent generated power higher than 2kW.

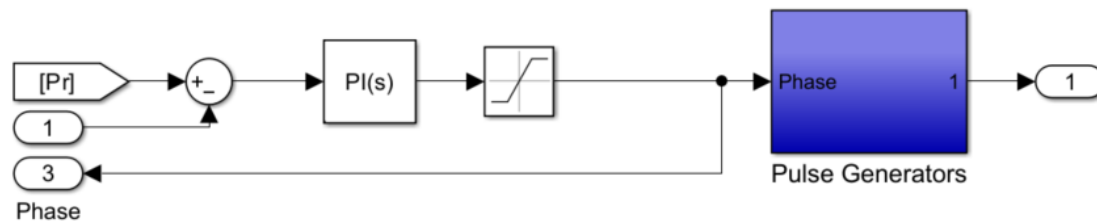


Figure 6: The control system of the DAB converter

After calculating the required phase shift it goes to the pulse generators block, [Figure 7](#) shows the pulse generation system for controlling the gates signals. It takes the phase shift signal and compares it with sawtooth signals, to determine the PWM signal characteristics, such as duty cycles. The output goes to flip-flops to stabilize and synchronize the signals, ensuring proper timing and avoiding overlapping states for safe operation. Transistors 1 and 4 must have the same PWM signal to work at the same time to provide a path to +V while transistors 2 and 3 provide a path to -V. This can be achieved by using NOT gates to invert the signals and ensure complementary switching for the transistors that must not work simultaneously.

Finally, the processed signals are routed to the gates, controlling the on and off states of the transistors, which control power transfer between the primary and secondary sides of the transformer. This system ensures precise timing and efficient operation of the DAB converter.

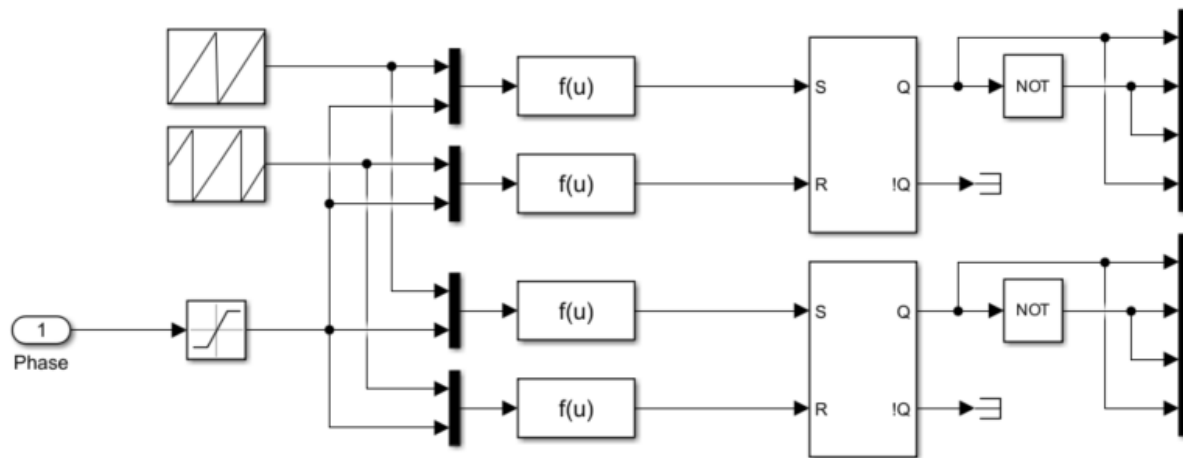


Figure 7: Pulse Generators Block

The Measurement and sensing system consists of Power measurement subsystem, Scopes to plot the outputs and efficiency calculator function. The power measurement block multiplies the current with the voltage to calculate the power then it divides it by 1000 to convert the result to kW.

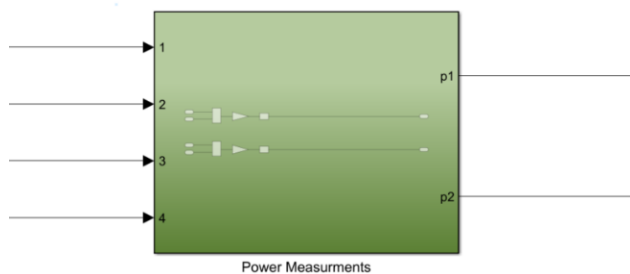


Figure 8: Power Measurements subsystem

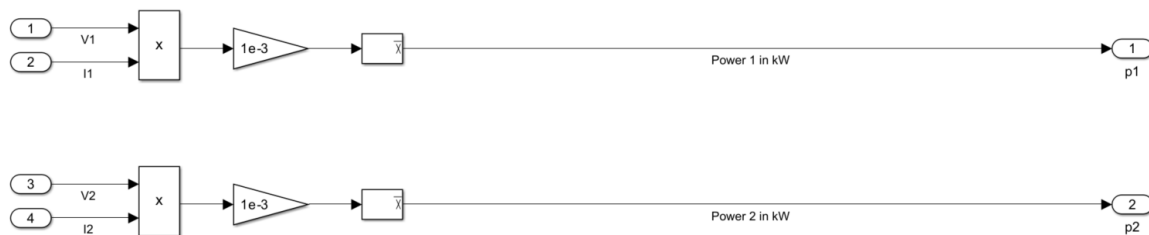


Figure 9: Power Calculator

Figure 10 illustrates the efficiency calculator block, it divides the lower power by the highest one depending on which side is discharging, then it multiplies the results by 100 to convert the number to 100%. To calculate the load percentage the calculator takes power and divides it by 2 then multiply it by 100.

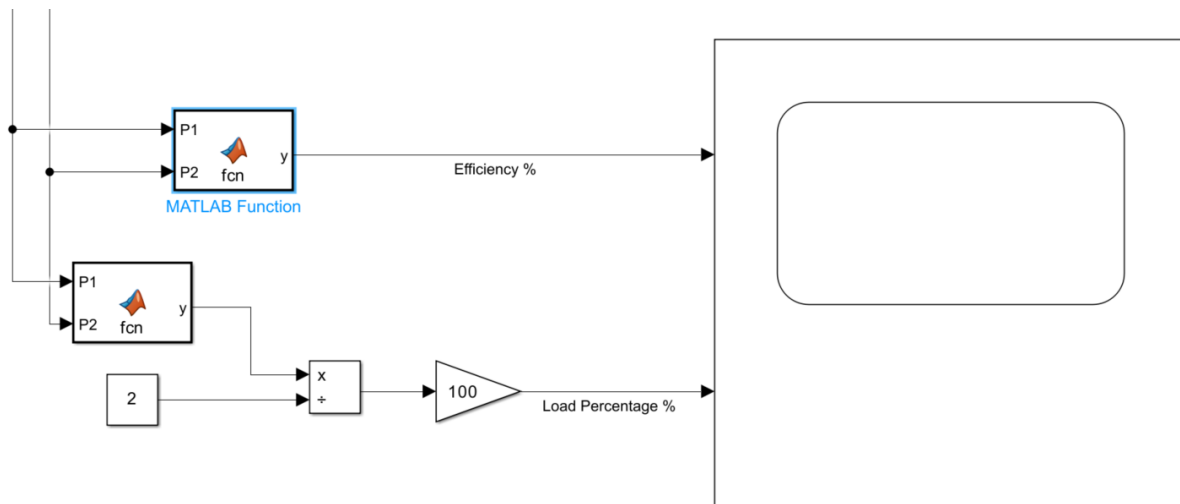


Figure 10: Efficiency Calculator

```

function y = fcn(P1, P2)

    if P1 == 0 && P2 == 0 %% to avoid undifiened statues
        y = 0;
        return;
    end

    if P1 == 0
        y = 0;
        return;
    elseif P2 == 0
        y = 0;
        return;
    end

    x = abs(P2 / P1);

    if x > 1
        y = (1/x) * 100;
    else
        y = (x) * 100;
    end
end

```

Figure 11: Software for Efficiency Calculator

```

function y = fcn(P1,P2)

if abs(P1)>= abs(P2)
y = abs(P2);

else
y=abs(P1);

end

```

Figure 12: Software for load percentage Calculator

## Simulation Results:

*Figure 13* shows the phase angle and the current curves over time, the phase angle increases until the current reaches the full load status which is around 44 amps then it saturates, in the discharging status the angle becomes negative which means negative current values. In a comparable way the power on the low voltage side starts increasing trying to follow the reference power until 2kW, then during the discharging case the power flows in the opposite direction which explains the negative power in the curve. In contrast, the high side voltage starts discharging first and then charging again, as shown in *Figure 14*.

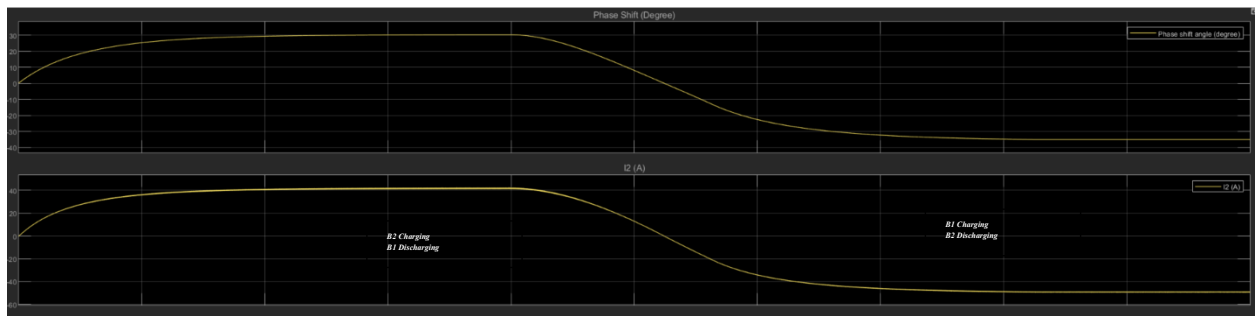


Figure 13: phase angle and current curves

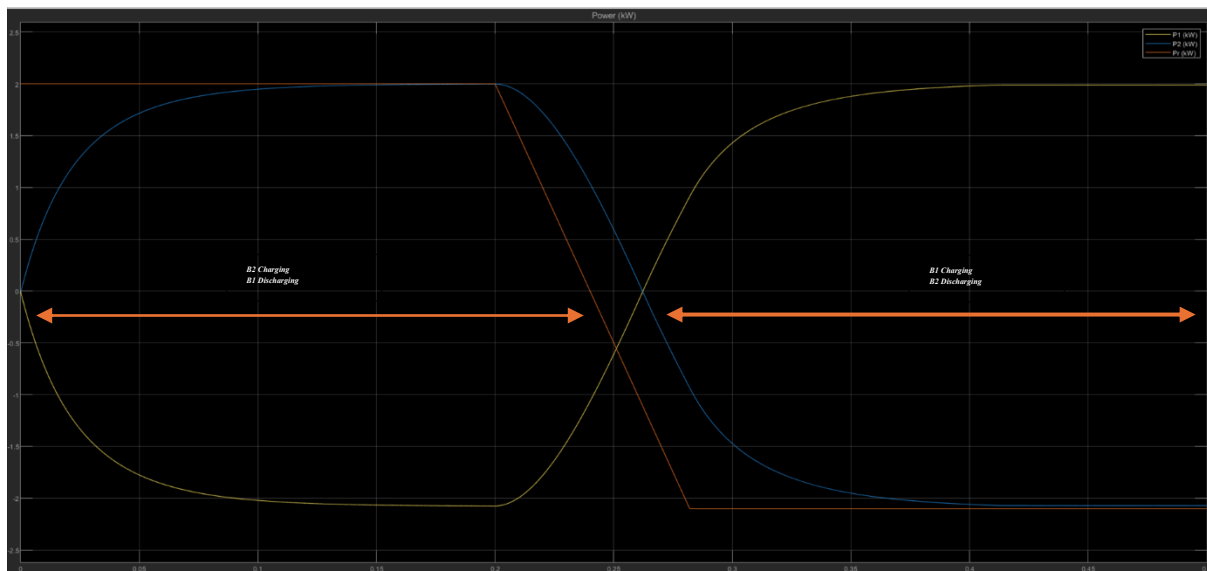


Figure 14: Input and output power curves

During the steady state, the efficiency is always higher than 95% which means that the converter's performance meets the project requirements in terms of power efficiency. Moreover, the results show that the voltage ripple is less than 3%.

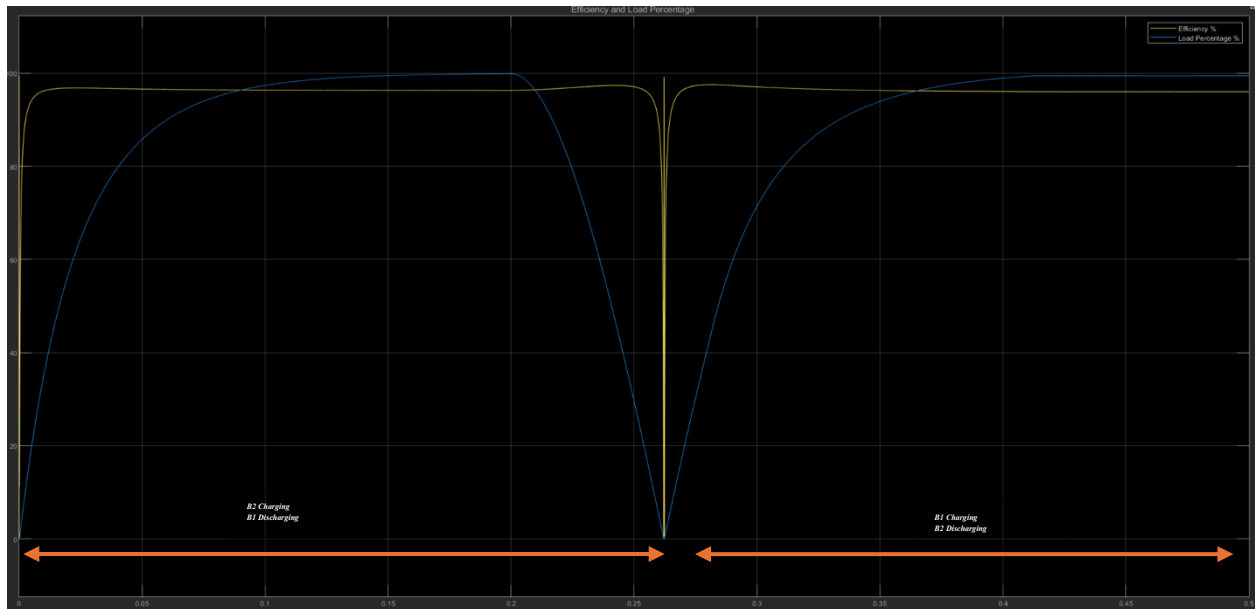


Figure 15: Efficiency curve

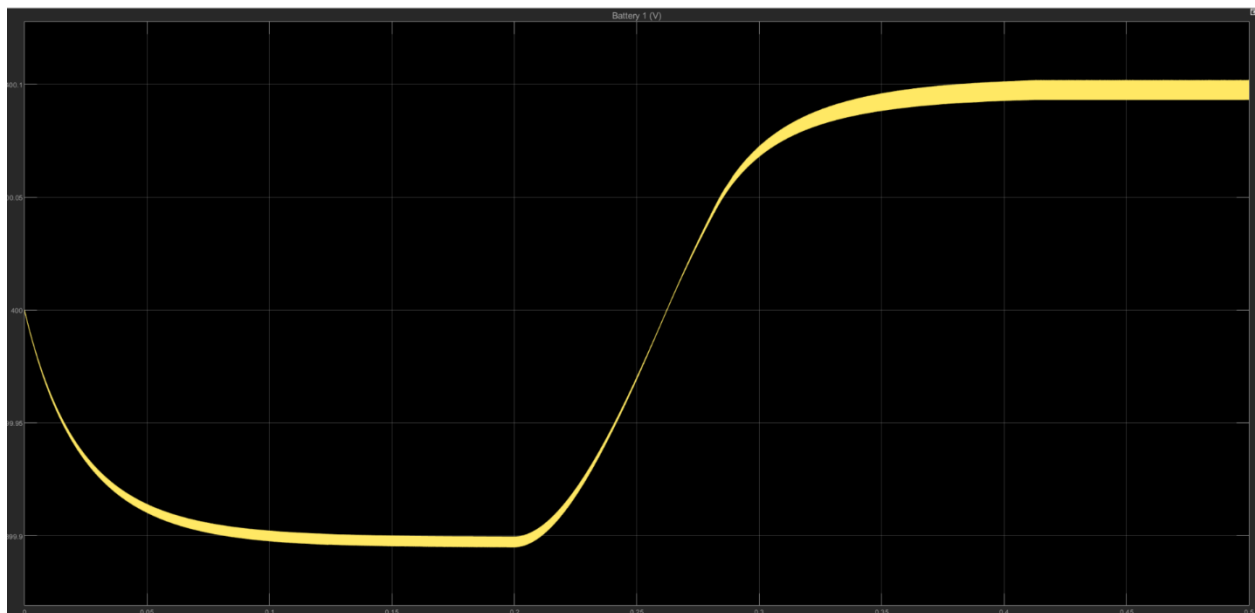


Figure 16: Battery Voltage Curve

## Discussion of Results:

The simulation showed a good result that met the project requirements, *Table 3* shows the project requirements and compares them with the results.

| Project Requirements                      | Simulation Results |
|-------------------------------------------|--------------------|
| Current Ripple <4% @ Full load (Charging) | <b>2.4%</b>        |
| Voltage Ripple <3% @ Full load            | <b>0.025%</b>      |
| Efficiency <95% @ Full load (Charging)    | <b>97%</b>         |
| Efficiency <92% @ 50 load (Charging)      | <b>96.7%</b>       |
| Efficiency <92% @ 25 load (Charging)      | <b>95%</b>         |
| Efficiency <95% @ Full load (discharging) | <b>98%</b>         |
| Efficiency <92% @ 50 load (discharging)   | <b>97.5%</b>       |
| Efficiency <92% @ 25 load (discharging)   | <b>97%</b>         |

Table 3: Project requirements vs results

*Figure 17* shows the current ripple, Equation 4 can be used to calculate the ripple Equation. 2.4% ripple current considered as a good percentage and met the specifications.

$$\%I_{\text{ripple}} = \frac{I_{\text{max}} - I_{\text{min}}}{I_{\text{mean}}} * 100\% \rightarrow \frac{42.3 - 41.3}{41.7} * 100 = 2.4\% \text{ (Equation 4)}$$

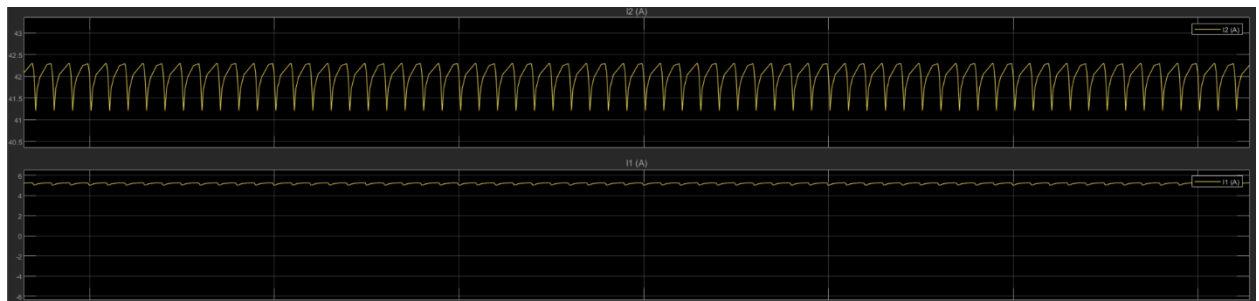


Figure 17: Current ripple curve



*Equation 5* shows how to find the voltage ripple, the ripple is always less than 3% in both discharging and charging statues.

$$\%V_{ripple} = \frac{V_{max}-V_{min}}{V_{mean}} * 100\% \text{ (Equation 5)}$$

$$\%V_{ripple-discharging} = \frac{399.95-399.85}{399.889} * 100 = 0.025\%$$

$$\%V_{ripple-charging} = \frac{400.12-400.09}{400} * 100 = 0.007\%$$

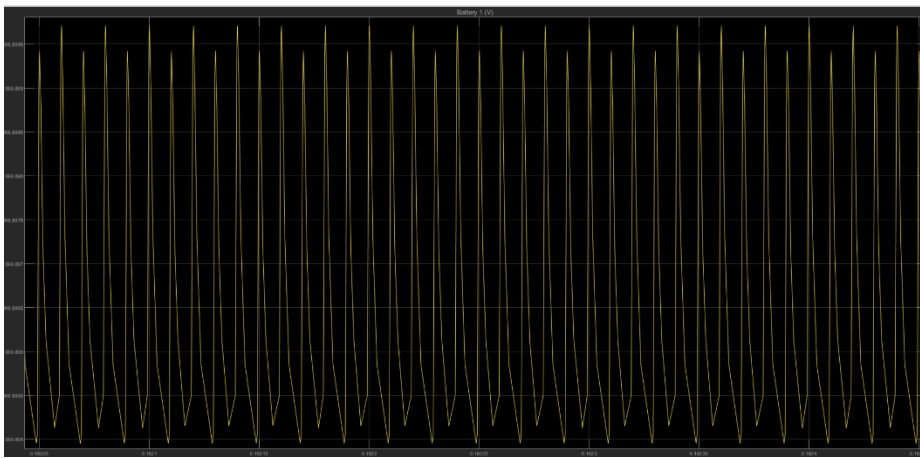


Figure 18: Voltage ripple during discharging

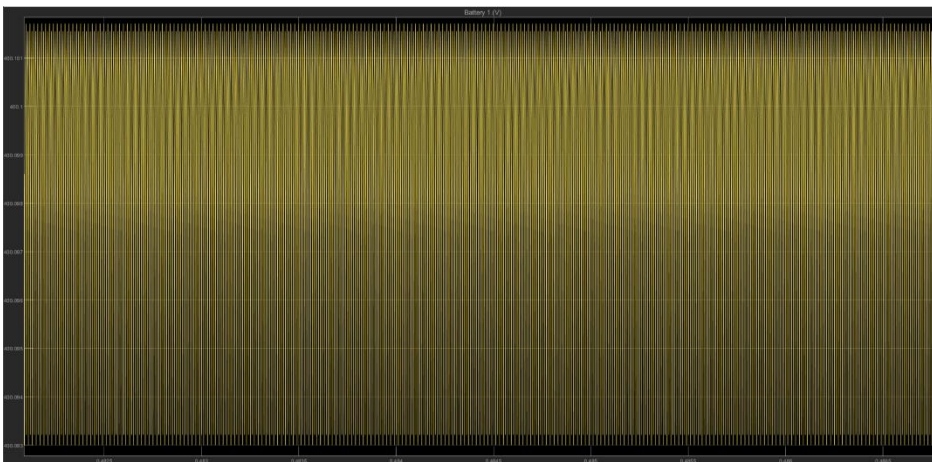


Figure 19: Voltage ripple during charging

According to the efficiency curve that shown in *Figure 20*, the efficiency is always higher than 95% at 25% load percentage during charging and discharging statuses. At full load, the efficiency equals 97%, using *Equation 6* the  $P_{loss}$  equals 0.0618 kW. These losses can be categorized into conduction loss, switching loss and transformer loss. Table 4 illustrates the loss types and their calculations.

$$\%Efficiency = \frac{P_{out}}{P_{loss}+P_{out}} * 100\% \quad (Equation\ 6).$$

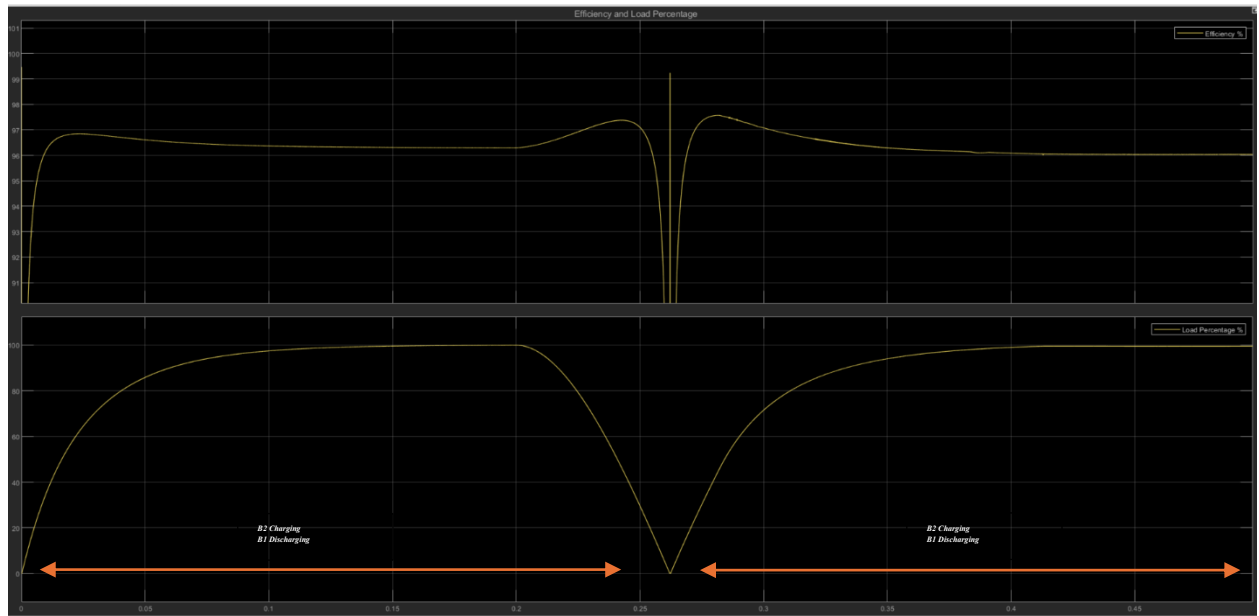


Figure 20: Efficiency vs Load percentage

| Power Loss                                       | Equation                          | Value in W @ Full load |
|--------------------------------------------------|-----------------------------------|------------------------|
| Conduction Sources Loss                          | $I^2 * R_{source}$                | 11.1                   |
| Conduction loss in transformers (primary side)   | $I^2 * R_{primary}$               | 0.25                   |
| Conduction loss in transformers (secondary side) | $I^2 * R_{secondary}$             | 0.0018                 |
| Transformer core loss                            | $K * B^{\beta} * f   I_m^2 * R_m$ | 3.16                   |
| MOSFET Conduction Loss                           | $I^2 * R_{on}$ for each on mosfet | 19                     |
| Diode Conduction Loss                            | $I^2 * R_d + I * V_d$ for each    | 22                     |
| Mosfet Switching Loss                            | $0.5 * V * I * f * T_{switching}$ | 2.5                    |
| Conduction loss in coupling inductor             | $I^2 * R_L$                       | 0.112                  |
| Conduction loss in DC link capacitor             | $I_{ripple}^2 * R_c$              | 0.0026                 |
| Total Loss                                       | $\sum P_{loss}$                   | 60                     |

Table 4: Loss breakdown calculations

```

% 1. Conduction loss in transformer (primary side)
P_cond_primary = I1^2 * R_primary;

% 2. Conduction loss in transformer (secondary side)
P_cond_secondary = I2^2 * R_secondary;

% 3. Core loss in transformer
P_core = k * (B_peak^beta) * f;

% 4. Conduction loss in Transistor
P_MOSFET = I1^2 * 2 * Ron + I2^2 * 2 * Ron;
P_diode = I1^2 * Rd + I2^2 * Rd + I2 * Vd;

% 5. Switching loss in MOSFETs
P_switch = 0.5 * V1 * I1 * f * t_switch;

% 6. Conduction loss in coupling inductor
P_inductor = I1^2 * R_coupling;

% 7. Capacitor ESR loss (primary side)
P_ESR_primary = I_ripple^2 * R_C;

% 8. Capacitor ESR loss (secondary side)
P_ESR_secondary = I_ripple^2 * R_C;

% 9. Conduction source loss
P_Rin = I1^2 * R1 + I2^2 * R2;

% Total Loss
Total_Loss = P_cond_primary + P_cond_secondary + P_core + P_MOSFET + ...
            P_switch + P_inductor + P_ESR_primary + P_ESR_secondary + P_diode + P_Rin;

% Efficiency
P_out = P_load;
Efficiency = (P_out / (P_out + Total_Loss)) * 100;

```

```

Loss Breakdown (W):
Conduction Loss (Primary Side): 0.2500 W
Conduction Loss (Secondary Side): 0.0018 W
Core Loss: 3.1623 W
MOSFET Conduction Loss: 19.6790 W
Diode Conduction Loss: 22.6101 W
MOSFET Switching Loss: 2.5000 W
Coupling Inductor Loss: 0.1125 W
Capacitor ESR Loss (Primary Side): 0.0026 W
Capacitor ESR Loss (Secondary Side): 0.0026 W
Batteries Conduction Loss: 11.0040 W
Total Loss: 59.4040 W
Efficiency: 97.12%

```

Figure 21: Loss breakdown calculator in MATLAB

**Conclusion:**

The simulation model of the proposed high-efficiency dual active bridge (DAB) converter proves its suitability and efficiency for distributed energy storage applications. The model gives an efficiency greater than 95% under all load conditions, with minimal voltage and current ripples, meeting the design requirements.

The implementation of a PI controller effectively controlled the phase shift angle, ensuring stable and precise power transfer between the high and low voltage sides. Loss analysis shows that the losses across components are minimal, which verifies the converter high performance and reliability. This simulation model represents a foundation for practical implementation in systems requiring efficient and safe energy distribution such as Electrical Vehicles and Smart Grids applications.

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